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ACADEMIC & LEADERSHIP EXPERIENCE	Teaching Assistant Classical Mechanics, under Professor Yu-tin Huang	2025.09–2025.12
	Research Assistant Theoretical Physics, under Professor Yu-tin Huang	2025.05–present
	Host MAKENTU Makerthon Contest	2025.05
	Head of R&D Department NTU Learning Optimization Club	2024.02–2024.06
COURSE PROJECTS	<u>Brief Introduction to Integrability and Thermodynamic Bethe Ansatz</u>	
	<i>Statistical Physics (II)</i>	Fall, 2025
	Introduced quantum integrability using the Lieb–Liniger model of a 1D interacting Bose gas, and then used the thermodynamic Bethe ansatz to study the thermodynamic properties, solving it numerically to explore interaction effects on momentum and energy distributions at finite temperature.	
	<u>A Brief Introduction to Black Hole Entropy</u>	
	<i>Advanced Topics in Gravity</i>	Spring, 2025
	Introduced two methods for calculating the entropy of Schwarzschild black holes and discussed the concept of microstates.	
	<u>Anisotropic Transmission of Quantum Information through Quantum Fields</u>	
	<i>Quantum Information and Computation</i>	Spring, 2025
	Studied quantum information transmission, combining the non-perturbative quantum field theory with the Unruh-De Witt model, and tried some approaches to the anisotropic transmission.	
	<u>A Brief Introduction to the Effective Theory of Binary Inspirals</u>	
	<i>Special Topics on Effective Field Theory and Scattering Amplitudes</i>	Fall, 2024
	Introduced the EFT framework, the hierarchy of scales, and the power counting rules for binary inspirals, and calculated the gravitational potential using Feynman diagrams.	
LANGUAGES & SKILLS	<u>A Brief Introduction to Conformal Bootstrap</u>	
	<i>Quantum Field Theory (II)</i>	Fall, 2024
	First introduced the basic concepts of conformal group, and then explained the algorithm of primary spectrum from the OPE associativity in Euclidean signature.	
	Languages: English (TOEFL iBT 106 : 29/28/22/27), Chinese	
	Programming: Mathematica, ROOT, MadGraph, Python, C++, L ^A T _E X.	
	Basic: Quantum Field Theory I & II (A+), Classical Mechanics (A+), Electromagnetics I & II (A+), General Relativity (A), Quantum Mechanics I (A+), Statistical Physics II (A+).	
RELEVANT COURSE-WORK	Advanced: Special Topics on Effective Field Theory and Scattering Amplitudes (A+), Quantum Information and Computation (A+), Advanced Topics in Gravity (A+), Basics of String Theory: from conformal field theory to supersymmetry (A+), Quantum Fields in Cosmology (A).	